

The markedness of geminate approximants and its perceptual grounding

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1. Four goals

1. To point out that geminate approximants are cross-linguistically marked (§2).
2. To argue that there is perceptual grounding for this markedness (§3).
3. To report an experiment that supports this hypothesis (§4).
4. To argue against the position that phonological patterns are shaped only by misperception (Ohala 1981 *et seq.*) (§5).

2. The markedness of geminate approximants

2.1. Japanese

- ☞ Geminate approximants are absent in the inventory.
- ☞ The creation of geminate approximants is avoided by coda nasalization (Kuroda 1965: 205).

- (1) /bata+μ+ri/ → [battari] ‘accidentally’
/poka+μ+ri/ → [pokkari] ‘opening a big hole’

- (2) *rr* → *nr*
/kira+μ+ri/ → [kinrari] ‘shiningly’
/horo+μ+ri/ → [honrari] ‘falling’

- ww* → *nw*
/yawa+μ+ri/ → [yanwari] ‘softly’
/huwa+μ+ri/ → [hunwari] ‘fluffy’

- yy* → *ny*
/doyo+μ+ri/ → [donyori] ‘cloudy’
/poyo+μ+ri/ → [ponyori] ‘chubby’

(3) *GEMAPPROX at work

/bata+μ+ri/	*GEMAPPROX	DEP(NAS)
a. ☞ battari		
b. bantari		*!
/kira+μ+ri/	*GEMAPPROX	DEP(NAS)
a. kirrari	*!	
b. ☞ kinrari		*

☞ It can affect the position of emphatic gemination (Kawahara and Akashi in prep.).

(4) C2 gemination when C2 is a non-approximant

/pata-pata+μ/ → [patta-pata] ‘fanning’
/kasa-kasa+μ/ → [kassa-kasa] ‘dried’

(5) C3 gemination when C2 is an approximant

/pura-pura+μ/ → [purapp-pura] ‘hanging around’
/poyo-poyo+μ/ → [poyopp-poyo] ‘soft’

(6) *GEMAPPROX at work II

/pata-pata+μ/	*GEMAPPROX	μ-to-σ ₁
a. ☞ patt <u>a</u> -pata		
b. patap <u>p</u> -pata		*!
/pura-pura+μ/	*GEMAPPROX	μ-to-σ ₁
a. purr <u>a</u> -pura	*!	
b. ☞ purap <u>p</u> -pura		*

2.2. LuGanda

(7) LuGanda gemination caused by class 5 prefix (Clements 1986: 62)

/kubo/ → [kkubo] ‘path’
/tabi/ → [ttabi] ‘branch’
/daala/ → [ddaala] ‘step’

(8) Occulusivization of approximants (Clements 1986: 68)

ll → dd
/μ-langa/ → [ddaanga] ‘lily’

yy → jj
/μ-yinga/ → [jjiinga] ‘stone’

ww → gg^w
/μ-wanga/ → [gg^waanga] ‘nation’

☞ Geminate approximants undergo hardening; hence *GEMAPPROX » ID (CONT), ID (SON).

(9) *GEMAPPROX at work III

/μ+langa/	*GEMAPPROX	IDENT(CONT)	IDENT(SON)
a. ☞ ddaanga		*	*
b. llaanga	*!		

2.3. Sanskrit (Whitney 1889)

(10) Geminate [r], a retroflex untrilled liquid, are resolved by degemination: /r+r/ → [r].

/punar <u>r</u> + <u>r</u> amate/	→	[puna: <u>r</u> amate]	(179)
/nṛpati <u>r</u> + <u>r</u> aṇati/	→	[nṛpati: <u>r</u> aṇati]	(179)
/ma:tu <u>r</u> + <u>r</u> ihaṇ/	→	[ma:tu: <u>r</u> ihaṇ]	(179)

☞ There is a clear prohibition against a geminate [r].

(11) Other types of geminates are admitted (often as a result of assimilation).

[tan- n amas]	(161)	[asse]	(166)
[ar ṇṇ a]	(189)	[atti]	(159)
[tal-labhate]	(162)	[arcad- d^h u:ma]	(159)
		[saccarita]	(202)
		[ya:taya ḥḥ ana]	(202)
		[ark k ka]	(228)
		[dig- g aya]	(159)

☞ The existence of geminate glides is far from clear: no clear examples of geminate glides in Whitney (1889).

☞ Whitney mentions a “context-free gemination process after [r],” which apparently creates [yy]: [ka:ryya] (228). However, Whitney also says that there are “authorities who deny it [the existence of such a gemination process] altogether.”

☞ Whitney identifies a historical stage in which degemination of glides was motivated.¹

☞ Geminate glides thus seem to be prohibited in Sanskrit as well. At the very least, we can motivate *[rr]:

(12)

/punar+ramate/	*[rr]	UNIFORMITY
a. ☞ puna:-ramate		*
b. punar-ramate	*!	

¹ This is based on “apparent loss (perhaps conversion to a semivowel) of i and u before y and v [=w], respectively” (233); e.g. /pari-yan/ → [payan], which underwent an intermediate stage where degemination is motivated; i.e., /pari-yan/ → [par**y**-yan] → [pa**y**an].

2.4. Other cases

	Alternations	Remarks	References
Berber	RR → qq ww → gg ^w	[ll] is not avoided.	Elmedlaoui (1995: 194-195)
Faroese	ww → gv yy → ʃʃ	Sonorant clusters pattern w/ geminate approximants.	Anderson (1972: 9-11)
Fula	ww → bb yy → ʃʃ rr → dd ²	Geminate fricatives are avoided as well.	Paradis (1992: 172-173), Keer (1999: 83-84)
Malayalam	rr → tt	Geminate glides are allowed.	Mohanan and Mohanan (1984: 581-582)
Chaha	*rr > nn *ll > nn	Historical process	Polotsky (1951: 36), McCarthy (1986: 22)
Endegeñ	rr → nn ll → nn ww → pp (?) ³	/xx/ → [kk] as well.	Leslau (1976: 142-143), Rose (2003)
Ezha	rr → nn		Leslau (1992)

Table 1: Other cases of avoidance of geminate approximants.

2.5. Finer distinctions

Languages can disallow only a subset of geminate approximants:

- (i) Berber allows [ll] but not [ww]. Sanskrit allows [ll] but not [rr].
- (ii) Finnish (Sherer 1994), Nubian (Bell 1971: 119-120, 127-133), and Trukese (Curtis 2003: 23) allow geminate liquids, but not geminate glides.
- (iii) Cuna disallows only [ww]; [yy] and all other geminate consonants are possible (Sherzer 1975: 49).
- (iv) Malayalam allows geminate glides but not [rr].

*GEMAPPROX = *GEMLIQUID, *GEMGLIDE
(=[*rr],*[ll]) (=[*ww],*[yy])

☞ I am not claiming that geminate approximants always behave as a natural class; I am claiming that they *can* form a natural class.

☞ Among geminate approximants, there is an implicational markedness hierarchy: [ll] seems always less marked than glide geminates; i.e. *GEMGLIDE » *[ll].

² Although Paradis (1992) gives no example of [rr]-hardening, she explicitly states that [rr] is prohibited.

³ Leslau treats this alternation as lenition of a singleton /p/ into [w] in intervocalic positions.

3. Phonetic grounding of *GEMAPPROX: A hypothesis

- ☞ The most important perceptual correlate of a geminacy distinction is constriction duration (Esposito and De Benedetto 1999; Hankamer et al. 1989; Lahiri and Hankamer 1988).
- ☞ Approximants have inherent disadvantages in signaling their constriction duration, and hence a constriction duration difference between singletons and geminates is difficult to perceive.
- ☞ Three reasons for why constriction duration is not very perceptible for approximants:
 - (i) Their segmental boundaries are blurry.
 - (ii) They involve smaller amplitude change in and out of flanking vowels.
(The steeper the amplitude change, the easier the segmentation is: Kato et al. 1997).
 - (iii) Their cues tend to extend over a domain containing larger than adjacent segments.
(It is hard to pin down where an approximant segment is: Ohala 1993; Blevins and Garret 2004).

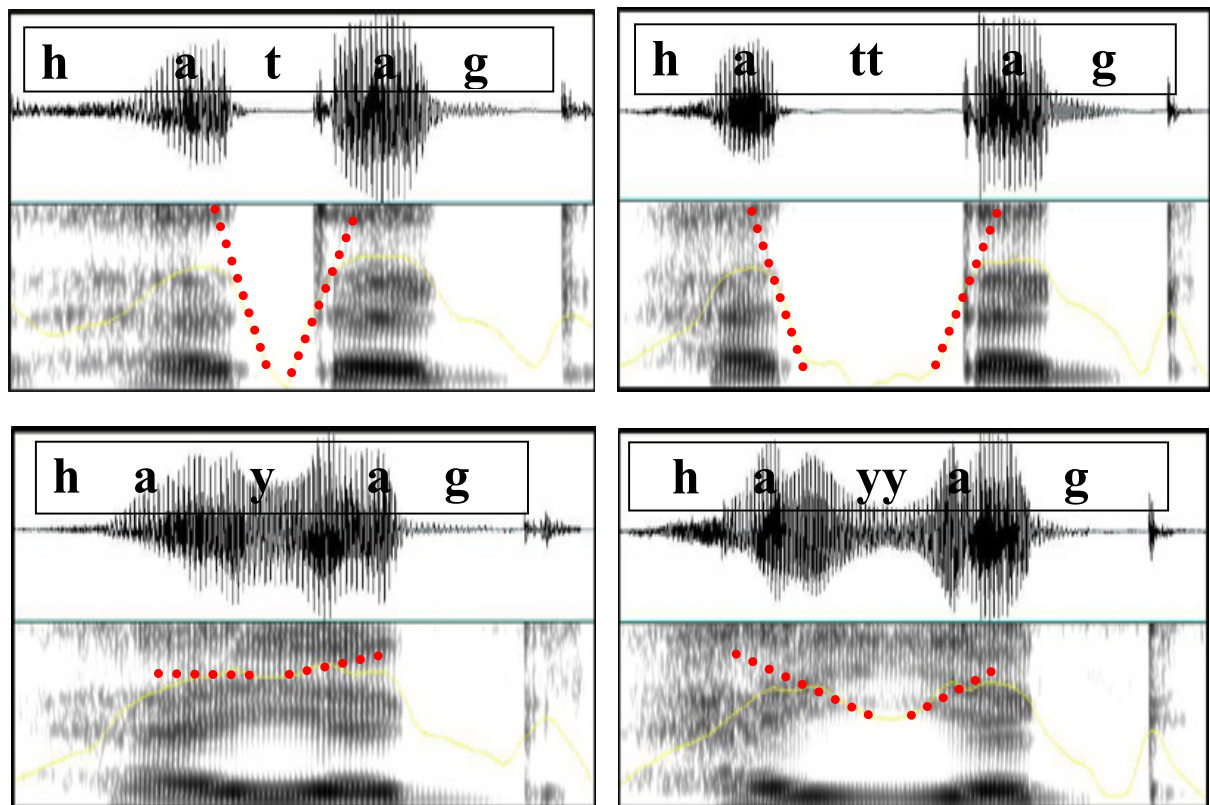


Figure 1: Spectrograms and waveforms of four nonce Arabic words, pronounced by a native speaker of Egyptian Arabic. Dotted lines represent the amplitude changes in and out of flanking vowels.

N.B: Appendix I illustrates spectrograms of other consonants; Appendix II shows numerical comparison of amplitude changes.

- ☞ This hypothesis has an immediate virtue of explaining *GEMGLIDE » *[ll], because the perceptual confusability problem is likely to be worse for glides than [ll].

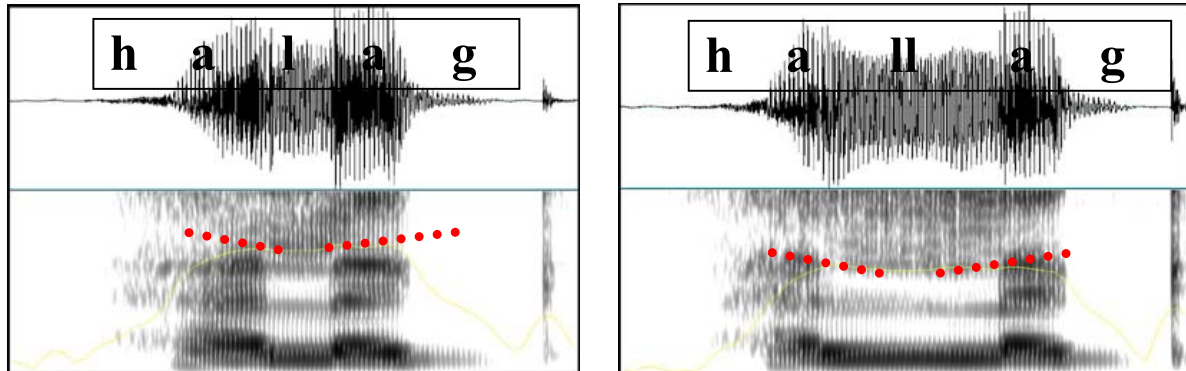


Figure 2: Spectrograms of singleton [l] and geminate [ll] in Arabic.

4. Experiment

4.1. Method

- ☞ A native speaker of Egyptian Arabic pronounced 14 nonce Arabic words:

- (13) hatag - hattag hasag - hassag hanag - hannag halag - hallag
 hadag - haddag hazag - hazzag hayag - hayyag

- ☞ Continua from singletons to geminates for each type of consonant were created, by repeatedly cutting off constriction duration from geminate endpoints:

→ 10 steps; each step was about 12ms.

→ Two kinds of continuum for each consonant: one with the preceding vowel which originally preceded a singleton token, and one with the preceding vowel which originally preceded a geminate token. (This turned out to have only a negligible effect, so this difference is averaged over in this talk).

- ☞ 140 stimuli in total (7 consonant types × 10 consonant lengths × 2 vowel lengths).

- ☞ 17 native speakers of Arabic heard the continua. They were from different dialects (Egyptian, Moroccan, Lebanese), but all dialects have all kinds of geminates, so the dialectal difference was not expected to impact the results.

→ An identification task where they were asked to judge the geminacy of the second consonant.

→ Each subject heard 15 repetitions of each token. All of the types of tokens were mixed together, and the order of presentation was completely randomized.

→ The data taken from three bilingual speakers were excluded from further analysis.

4.2. Results

4.2.1. Identification functions

Prediction:

If listeners are more confused in figuring out a geminacy distinction of approximants, then the identification function should have a shallower slope.

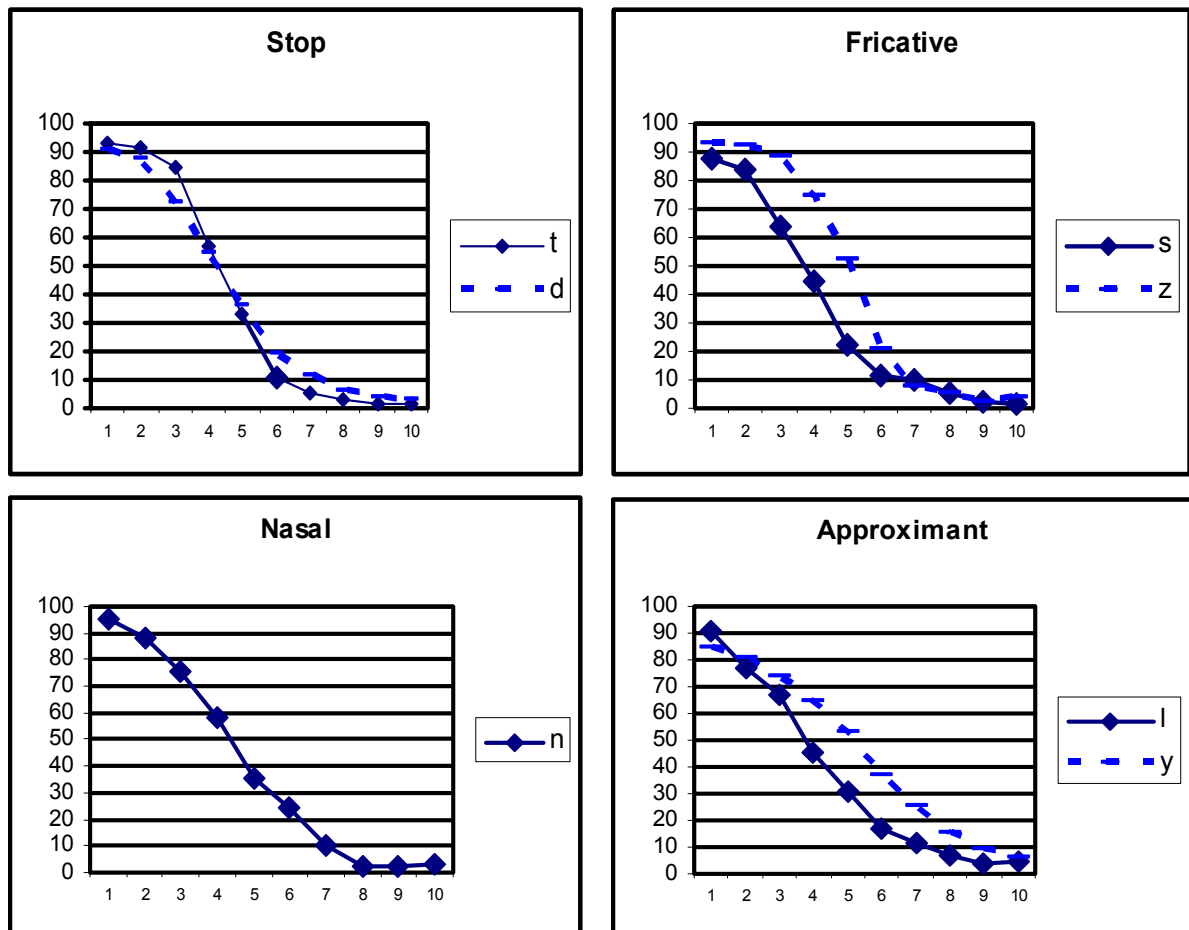


Figure 3: Identification functions for the seven types of consonants. The y-axis represents the proportion of singleton response across 15 repetitions; the x-axis represents 10 steps in constriction duration continua.

- ☞ Apparently, approximants have shallower and more linear function, especially compared to obstruents. Nasals seem to be in-between.

- ☞ To quantify the steepness of the slopes, the slope coefficient between *the last point above y=80* and *the last point above y= 10* was calculated for each consonant.

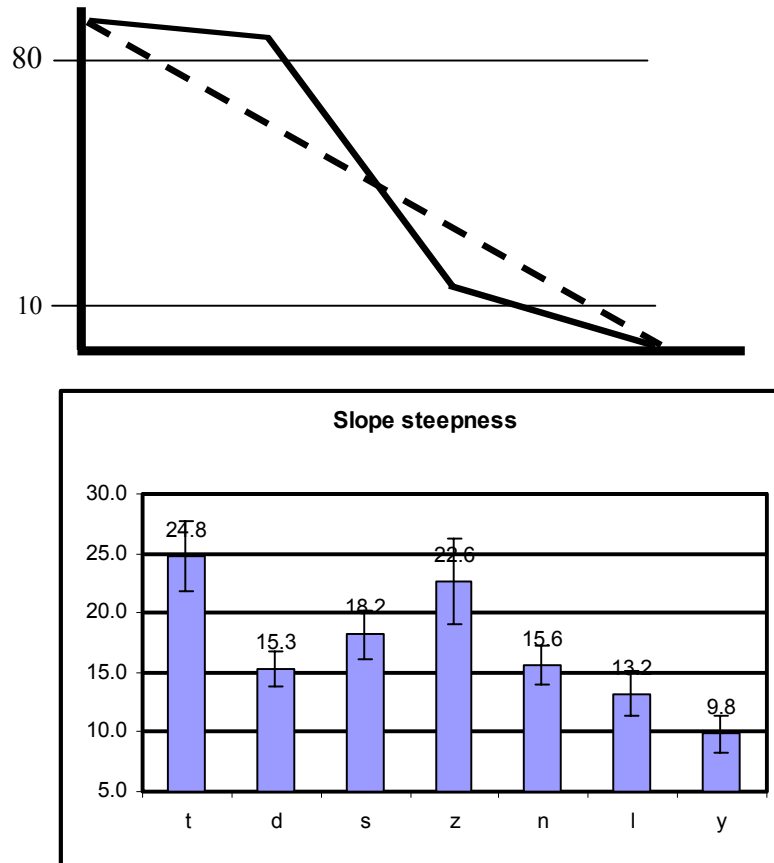


Figure 4: The slope coefficient for each type of consonants. The error bars represent 95% confidence intervals.

- ☞ Approximants, especially [y], have shallower slopes than other consonants ($p < .0001$).

- ☞ The slope for [l] is significantly different from that of [d] and [n] ($p = .003$).

- ☞ The slope of [y] is shallower than that of [l] ($p = .002$).

→ This indicates that it is more difficult to perceive a geminacy distinction for [y] than it is for [l]. This may be the cause of the ranking, *GLIDE GEM » *[ll].

N.B: Details of the statistical analyses are shown in Appendix III.

4.2.2. Reaction time

Prediction:

If it is more difficult to perceive a geminacy distinction for approximants, then reaction time should be longer for approximants than non-approximants.

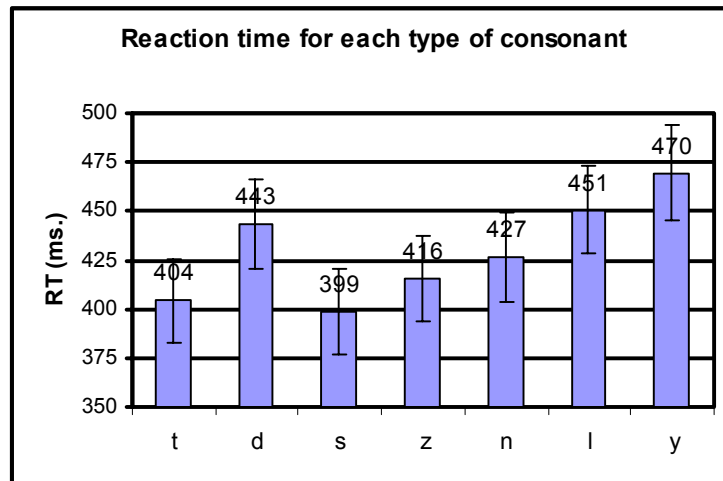


Figure 5: Reaction time for the seven types of consonants. Those which were longer than the mean + 2 × the standard deviation for each speaker were excluded as outliers.

- ☞ The difference between approximants and non-approximants is statistically significant ($p < .0001$)
- ☞ The reaction time for [l] is longer than the reaction time for [d] and [n] ($p = .045$).
- ☞ There is no evidence for a difference between [l] and [y] ($p = 0.408$).

Summary of the experiment:

A geminacy distinction is harder to discriminate in approximants than in non-approximants, as evidenced by:

- shallower identification functions
- longer reaction time

5. Summary and discussion

A. The cross-linguistic markedness of geminate approximants

☞ Geminate approximants are resolved by a variety of processes; it exhibits heterogeneity of processes: McCarthy 2002; Pater 1999):

(14) Heterogeneity of processes due to *GEMAPPROX

- coda nasalization (Japanese)
- flopping of a floating mora (Japanese)
- occlusivization (Berber, Fula, Malayalam, LuGanda)
- nasalization (Chaha, Endegeñ, Ezha)
- degemination (Sanskrit)

B. The perceptual grounding of this markedness

☞ A geminacy contrast of approximants is harder to discriminate compared to other consonants, which is likely to be the cause of the markedness of geminate approximants.

☞ Confusability of a geminacy distinction for approximants is phonologized as a constraint *GEMAPPROX, which can subsequently be satisfied by a range of phonological processes.

- This provides support for a model that incorporates functional grounding internal to the formal grammar (Archangeli and Pulleyblank 1994, Boersma 1998, Hayes 1999, Smith 2002, 2004, Steriade 2001 among many others)

C. Phonology is not just a matter of misperception

☞ This is *incompatible* with the “misperception + phonologization” approach to phonology (Barnes 2002; Blevins 2004; Blevins and Garrett 2004; Kavitskaya 2002; Ohala 1983, 1991).

- [x] is absent in language L because [x] was misperceived as [y], and diachronically became [y], or the misperception of [x] as [y] is “phonologized” as an alternation /x/ → [y].
- Thus, the prediction is that geminate approximants are *always* resolved by degemination, because geminate approximants are confusable with singleton approximants.

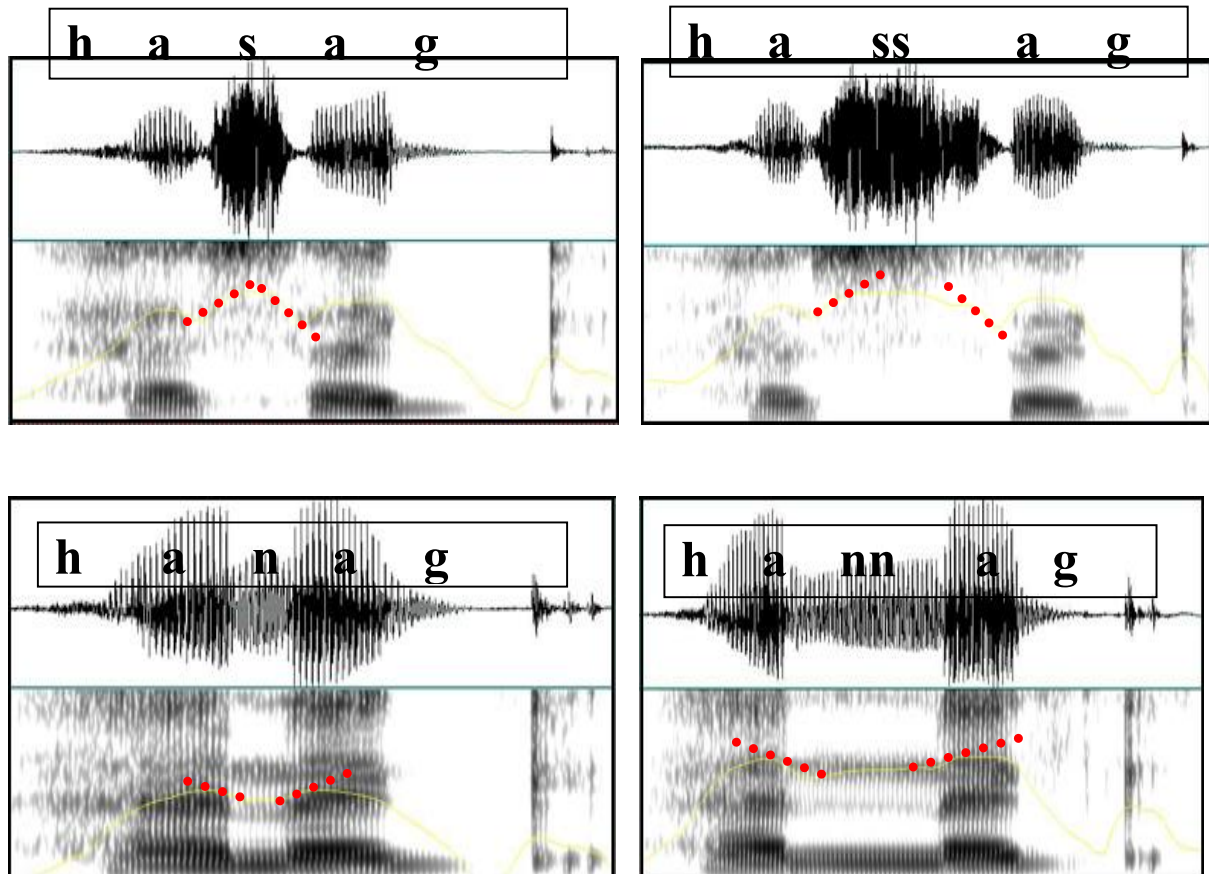
☞ However, this is inconsistent with the heterogeneity of processes in (14).⁴ Rather the confusability problem is phonologized as a constraint *GEMAPPROX, which can induce a variety of phonological processes.

⁴ Kingston (1999), Smith (2004) and Steriade (2001: 232-235) independently develop arguments against the misperception+phonologization approach to phonology.

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Appendix I: Spectrograms of nasals and fricatives in Arabic.



Appendix II: Amplitude changes

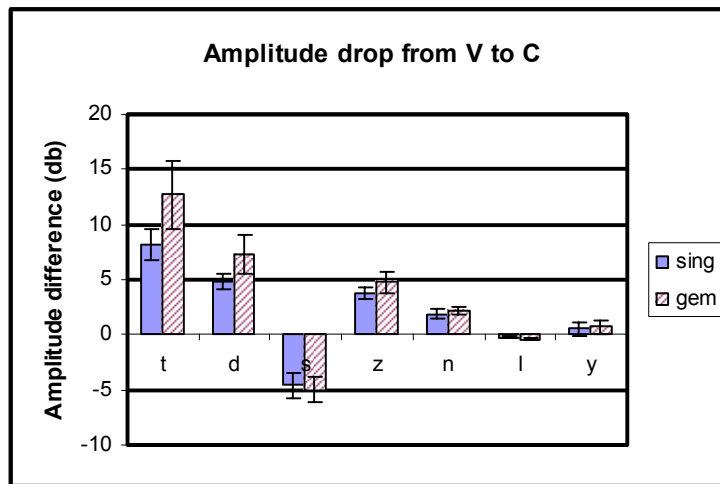


Figure 6: The amplitude drop in V-to-C transition, calculated based on 10 repetitions. Positive values represent an amplitude drop; negative values represent an amplitude rise. Error bars represent 95 percent confidence intervals.

The difference between approximants and non-approximants in the absolute magnitude of amplitude changes is statistically significant, $F(1, 136)=81.244$, $p<.0001$. Geminacy does not affect the size of amplitude change, $F(1, 136)=2.716$, $p=.102$.

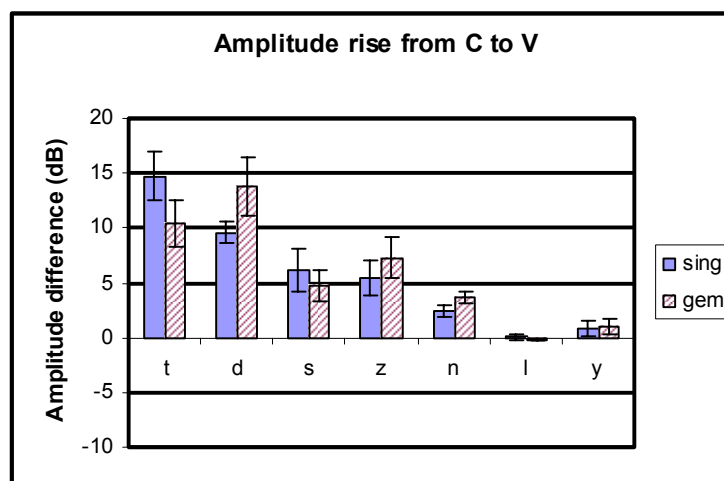


Figure 7: Amplitude rise in C-to-V transition: the amplitude value of the following vowel minus the amplitude of the right consonantal edge. Error bars represent 95 percent confidence intervals.

Again, the size of amplitude change is larger for non-approximants than for approximants, $F(1, 136)=91.821$, $p<.0001$. The effect of geminacy is not significant at all, $F(1, 136)<1$.

Appendix III: Statistical details for the experiment

All the analyses are two-tailed post-hoc within-subject contrast analyses. The alpha level is adjusted to $0.05/3=0.017$ by a Bonferroni FWE adjustment.

μ represents a population mean; ψ represents a contrast among μ s. For example, a contrast defined as $\psi=\mu_d-1/2(\mu_t+\mu_s)$ represents “the difference between the population mean of [d] and the average of the population means of [t] and [s].” ψ s and their standard errors were estimated by:

$$\hat{\psi} = \sum \sum w_j \bar{Y}_{ij} / n$$

$$S\hat{\psi} = [\sum \sum (w_j \bar{Y}_{ij} - w_j \bar{Y}_{.j})^2 / \{(n-1)*(n)\}]^{0.5}$$

$$t(n-1) = \hat{\psi} / S\hat{\psi}$$

(where $\bar{Y}_{ij}$ is the mean for i th subject of j -type segment; w_j is the weight associated with j -type segment such that $\sum w_j=1$; n is a number of subjects)

1. Slope coefficient

☞ *Approximants, especially [y], have shallower slopes than other consonants.*

$$\psi_1 = 1/5(\mu_t + \mu_d + \mu_s + \mu_z + \mu_n) - 1/2(\mu_l + \mu_y)$$

$$\hat{\psi}_1 = 7.8, t(13) = 10.04, p < .0001$$

☞ *The slope for [l] is significantly different from that of [d] and [n].*

$$\psi_2 = 1/2(\mu_d + \mu_z) - \mu_l$$

$$\hat{\psi}_2 = 2.2, t(13) = 3.72, p = .003$$

☞ *The slope of [y] is shallower than that of [l].*

$$\psi_3 = \mu_l - \mu_y$$

$$\hat{\psi}_3 = 3.4, t(13) = 3.91, p = .002$$

2. Reaction time

☞ *The difference between approximants and non-approximants is statistically significant.*

$$\psi_4 = 1/2(\mu_l + \mu_y) - 1/5(\mu_t + \mu_d + \mu_s + \mu_z + \mu_n)$$

$$\hat{\psi}_4 = 46.05, t(13) = 9.222, p < .0001$$

☞ *The reaction time for [l] is longer than the reaction time for [d] and [n].*

$$\psi_5 = \mu_l - 1/2(\mu_d + \mu_n)$$

$$\hat{\psi}_5 = 24.33, t(13) = 2.51, p = .045 \text{ (only barely significant)}$$

☞ *There does not seem to be a difference between liquids and glides.*

$$\psi_6 = \mu_y - \mu_l$$

$$\hat{\psi}_6 = 12.27, t(13) = 0.888, p = 0.408$$

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